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Face–name association learning in early Alzheimer's disease: A comparison of learning methods and their underlying mechanisms

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This study compared the efficacy of five learning methods in the acquisition of face–name associations in early dementia of Alzheimer type (AD). The contribution of error production and implicit memory to the efficacy of each method was also examined. Fifteen participants with early AD and 15 matched controls were exposed to five learning methods: spaced retrieval, vanishing cues, errorless, and two trial-and-error methods, one with explicit and one with implicit memory task instructions. Under each method, participants had to learn a list of five face–name associations, followed by free recall, cued recall and recognition. Delayed recall was also assessed. For AD, results showed that all methods were efficient but there were no significant differences between them. The number of errors produced during the learning phases varied between the five methods but did not influence learning. There were no significant differences between implicit and explicit memory task instructions on test performances. For the control group, there were no differences between the five methods. Finally, no significant correlations were found between the performance of the AD participants in free recall and their cognitive profile, but

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generally, the best performers had better remaining episodic memory. Also, case study analyses showed that spaced retrieval was the method for which the greatest number of participants (four) obtained results as good as the controls. This study suggests that the five methods are effective for new learning of face–name associations in AD. It appears that early AD patients can learn, even in the context of error production and explicit memory conditions.

Keywords: Alzheimer's disease; Cognitive rehabilitation; Face–name learning; Errorless learning; Implicit memory.

INTRODUCTION

Severe impairment in episodic memory is one of the core features of Alzheimer's disease (AD) (Collette, Van der Linden, Juillerat, & Meulemans, 2003; Hodges, 2000; McKhann et al., 1984). In the early phases of the disease, the episodic memory impairment is usually associated with deficits in attention and executive functions (Amieva et al., 2004; Collette, Van der Linden, & Salmon, 1999) and in semantic memory (Hodges, 2000; Perry & Hodges, 2000), although these deficits are generally less severe. However, many cognitive capacities are still preserved, more specifically in some aspects of memory such as perceptual and procedural memory (Deweer et al., 1994). This population is thus likely to benefit from a cognitive intervention consisting of optimising these preserved learning abilities. In fact, recent studies have shown the efficacy of cognitive rehabilitation in early AD aimed at helping patients, with learning methods, to learn or relearn specific relevant knowledge and abilities that will have a direct impact on autonomy (Adam, Van der Linden, Juillerat, & Salmon, 2000; Camp, Foss, Stevens, & O'Hanlon, 1996b; Clare, Wilson, Breen, & Hodges, 1999; Clare, Wilson, Carter, Hodges, & Adams, 2001). The main learning methods studied in AD are errorless learning, spaced retrieval, and vanishing cues.

The errorless learning method aims at minimising the production of errors. It consists of providing the participant with the correct answer and allowing the repetition of this answer several times (Baddeley & Wilson, 1994). This method rests on observations made by Tulving and collaborators (Hayman, Macdonald, & Tulving, 1993; Tulving, Hayman, & Macdonald, 1991) and Baddeley (Baddeley, 1992; Baddeley & Wilson, 1994) that amnesics have great difficulty avoiding previous mistakes. Since they have a severe episodic memory deficit, they cannot remember that they produced an error on the preceding learning trials. Furthermore, their preserved implicit memory, which refers to an automatic and unintentional retrieval of information that is not supported by a recollection of the learning episode (Schacter, 1995), will tend to produce the error automatically across trials (Baddeley & Wilson, 1994).

Thus, amnesics tend to repeat their mistakes over and over again, reinforcing them each time they are produced. The errorless method seeks to enhance learning by limiting the number of mistakes produced and ensuring that only the correct answer will be repeated. Studies in AD showed that this method is promising for face–name learning alone (Metzler-Baddeley & Snowden, 2005; Winter & Hunkin, 1999) or in combination with other methods, like imagery (Clare et al., 1999, 2000, 2001). However, recent studies indicate that the impact of limiting errors to promote learning in AD should be further explored.

Metzler-Baddeley and Snowden (2005) compared errorless with errorful conditions in four participants with AD. They were asked to learn object names as well as familiar and novel face–name associations. The authors found a significant advantage of errorless over errorful learning for the group but no advantage of errorless over errorful learning on an individual level. As for Haslam et al. (2006), they showed that, for several participants with AD or dementia of mixed diagnoses, there was no significant advantage of errorless over trial-and-error for learning face–name–occupation associations.

The limitation of errors as a way to promote learning in AD was also considered in the efficacy of the second learning method, spaced retrieval, which is the most developed and studied method in the AD population. In this method, the participant is asked to recall information after various increasing time intervals. At the beginning, the intervals are closer together to ensure that the patient has learned the information. Thereafter, the intervals are spaced according to the patient's performance: if he/she makes a mistake, the error is corrected immediately and the last successful interval is repeated, after which the time series is continued (Camp, Foss, O'Hanlon, & Stevens, 1996a). This method seems promising in AD for various tasks potentially useful in increasing functional autonomy, such as face–name learning (Camp & Stevens, 1990), prospective memory tasks like calendar use (Camp et al., 1996a; McKitrick, Camp, & Black, 1992), object names (McKitrick & Camp, 1993), and using a cell phone (Lekeu, Chicherio, Van der Linden, & Slamon, 2000). However, not all individuals seem to learn with this method but the specific cognitive profile of these participants is not clearly described (Camp et al., 1996b; Hochhalter, Bakke, Holub, & Overmier, 2004; Hochhalter, Overmier, Gasper, Bakke, & Holub, 2005).

The rationale for the use of spaced retrieval in AD is a matter of debate. As mentioned earlier, one of the hypotheses put forward in the literature is that spaced retrieval limits the amount of errors produced by the participants since the first intervals are very close together. Recent studies have shown that spaced retrieval does not limit error production more than other learning methods, such as random rehearsal and massed repetition, and shows the same efficacy (Hochhalter et al., 2005). Another hypothesis is the possible capacity of this method to tap preserved implicit memory capacities

(Camp, 1989; Camp et al., 1993; Cherry, Simmons, & Camp, 1999). As mentioned earlier, implicit memory refers to an automatic and unintentional retrieval of information. Implicit memory is measured via different tests in which the participant answers without knowing that his/her actual performance is influenced by mnemonic activities that took place during the learning phase. In these tests, administration is generally designed to avoid having participants establish a link between the training and the test phase by asking them to complete a task as fast as possible or give the first answer that comes to mind (Schacter, 1995). Conversely, explicit memory refers to a conscious and controlled retrieval of information (Schacter, 1995). Classic explicit memory tests include free recall, cued recall and recognition in which the subject is specifically asked to recollect the information that was provided earlier. Thus, one of the main differences between implicit and explicit memory tests lies in the instructions given to the subjects when they are tested.

The implicit memory hypothesis to explain the efficacy of the space-retrieval method has not been explored thoroughly in AD and is mainly based on anecdotal evidence. For example, it has been observed that some AD participants recall answers without effort or without remembering having encountered them before (Camp & McKitrick, 1992). More direct, but modest, evidence was obtained by Cherry and collaborators (Cherry & Simmons-D'Gerolamo, 1999; Cherry et al., 1999). Implicit memory contribution was measured via a category exemplar generation task used in pre- and post-intervention. After the spaced retrieval intervention, which consisted of learning to identify three target objects, half of the AD participants named one of these objects during the category exemplar generation task. However, many patients also named these objects during the pre-test and there was no comparison of the results with chance level.

The debate surrounding the implicit memory contribution to the spaced retrieval method has also been raised for the third learning method, vanishing cues. This method was mainly studied with amnesic patients and takes as a starting point traditional techniques of forward and backward chaining, which consist of graduating the assistance given to the participant. The vanishing cues method is also based on a lexical priming task (implicit memory test) in which the participants must complete the first letters of words with the first words that come to mind (Glisky, Schacter, & Tulving, 1986). The rationale was that amnesic patients obtain normal or near normal performances in these priming tasks. Glisky and colleagues (1986) thus tried to incorporate the characteristics of these implicit memory tests in a learning method. Generally, patients are provided with the first letter of a word to be memorised (S _ _ _) and must try to guess the correct answer (SAVE) (Glisky, Schacter, & Tulving, 1986). If they cannot find the answer, a letter is added (SA _ _) until they are able to give the right answer. In the following trial, the number of

cues given is one less than the number necessary in the preceding trial, and so on, until the word is recovered without any cues (Glisky & Schacter, 1987, 1988; Van der Linden & Coyette, 1995). In AD, this method was successfully used for learning face–name and face–occupation associations in two AD patients (Fontaine, 1995) and in combination with other strategies also for face–name association (Clare et al., 1999, 2000). The implicit memory contribution to the efficacy of vanishing cues in AD was explored in Fontaine's (1995) study by means of a priming task in which the two participants showed implicit memory for items learned with vanishing cues. The efficacy of vanishing cues has also been explained by its possible capacity to limit errors (Baddeley & Wilson, 1994). However, in AD, Fontaine (1995) observed that these participants produced many mistakes with the original procedure. An alternative to this procedure, which consisted of providing the answer first (SAVE) before gradually decreasing the cues given (SAV_; SA _ _), could potentially decrease the number of mistakes produced since the participant does not have to guess the answers during the initial trials (Fontaine, 1995; Hunkin & Parkin, 1995).

In sum, actual results have shown that errorless, spaced retrieval and vanishing cues methods are promising. However, many of their aspects have not been studied in early AD. First, some methodological considerations were not controlled in previous studies. As pointed out by Hochhalter and colleagues (2005), few studies included a control condition, such as other learning methods, to confirm the efficacy of the learning method tested. In addition, few details are given about the cognitive profile of the tested participants. Finally, there are very few comparisons of the three methods in the same patients, and the results of the studies are contradictory (Bourgeois et al., 2003; Clare et al., 2000; Hochhalter et al., 2004, 2005). Thus there are no clear guidelines for their clinical use.

Secondly, the underlying mechanisms of the three learning methods still need to be clarified in AD. Few studies have documented the number of errors produced with each of these methods or explored if error limitation is an important factor to promote new learning in this population. Also, the implicit memory assumption has not been studied extensively. The contribution of implicit memory could be explored in different ways. One possibility, which has not been explored to our knowledge, is to ensure that these methods do promote implicit memory by reproducing, in their procedures, the methodological conditions that prevail during implicit memory tests; for example, by asking the participants, during the learning phase, to complete the task as fast as possible or give the first answer that comes to mind. In fact, we think that the instruction that is used is some studies, which is to ask the participants not to guess if they are not sure of the answer (see Clare et al., 1999; Haslam, Gilroy, Black, & Beesley, 2006), does not optimise implicit memory and is not a way to avoid errors, since

the participants can still try to explicitly search in their memory to find the right answer. The patients' performance with the implicit instructions could also be compared to their performances with methods using explicit memory test instructions, which specifically ask participants to recall the information presented earlier.

Based upon these methodological and theoretical considerations, the first aim of this study was to explore if each target method – errorless, spaced retrieval and vanishing cues – is efficient in increasing learning of face–name associations in early AD. The second aim was to determine if one method is more efficient than the others by comparing them. We also explored the impact of error production on learning (third aim) by comparing the three methods with trial-and-error conditions, which are assumed to promote errors during learning. We also measured the amount of errors produced with each method and explored the impact of this error production on learning. The fourth aim was to explore the contribution of implicit memory to the methods' efficacy. To do so, we used implicit memory test instructions during the method's learning phase and compared them with a condition using explicit memory test instructions. Finally, we explored if some cognitive functions were associated with the participants' performances, in order to determine the characteristics of the best and worst performers.

METHODS

Participants

Two groups of participants took part in this study: participants with early AD ($n = 15$) and controls ($n = 15$) matched for age, sex and years of schooling. Demographic characteristics and general cognitive functioning of both groups are presented in Table 1. The AD participants met the NINCDS-ADRDA (McKhann et al., 1984) or DSM-IV criteria (American Psychiatric Association, 1994) for probable Alzheimer's disease or dementia of the Alzheimer type on the basis of a complete clinical evaluation (including geriatric and neuropsychological evaluations) carried out by a multidisciplinary team. Of the 15 patients, 8 had taken inhibitors of cholinesterase for at least 6 months. The exclusion criteria were: (1) cognitive deficits that could interfere with the procedure: comprehension deficit or significant visuoperceptual deficits that could limit recognition of faces; (2) major depression or anxiety; (3) behaviour problems; and (4) all other types of dementia (vascular, semantic, alcoholic, etc.).

All AD participants also received a standardised neuropsychological evaluation for the purpose of this study (the controls were not submitted to the evaluation). Results are shown in Table 2. The AD group as a whole

TABLE 1
Demographic characteristics of participants and level of general cognitive function

Characteristics	*AD* ($N = 15$) (Mean $\pm$ *SD*)	*Controls* ($N = 15$) (Mean $\pm$ *SD*)	*p* values
Age	73.3 $\pm$ 7.9	72.5 $\pm$ 7.9	.58
Sex (number of women)	9	9	-
Years of schooling	10.5 $\pm$ 3.7	11.7 $\pm$ 5.9	.51
MMSE (/30)	23.7 $\pm$ 3.2	29.2 $\pm$ .70	$<$.001
Mattis Dementia Rating Scale (/144)	122.4 $\pm$ 11.3	136.8 $\pm$ 6.9	$<$.001

SD = standard deviation. Mini-Mental State Evaluation (MMSE; Folstein, Folstein, & McHugh, 1975); Mattis Dementia Rating Scale (Mattis, 1976).

presented mainly episodic memory, semantic memory and inhibition deficits, in accordance with the general cognitive profile of early AD (Collette et al., 2003; Hodges & Patterson, 1995). Also, as a group, they obtained a score of 23.7 $\pm$ 3.2 on the Mini-Mental State Examination (MMSE), which is defined as minimal to mild AD (Hodges, 2000). Thirteen participants were recruited at the Research Center on Aging (Sherbrooke, Canada) and two further participants were recruited in Belgium (Memory Day Care Center, Liège).

The control participants were recruited by word of mouth and through advertisements. Exclusion criteria included all types of dementia, memory complaints, history of head injury, depression, medication that could influence memory performance (e.g., anticholinergic medication) and a Mattis Dementia Rating Scale abnormal for age and level of schooling. The project was approved by the Research Ethics Committee of the Research Center on Aging, Sherbrooke, Canada. Informed written consent was obtained from both AD and control participants.

Materials and procedure

Materials

The task comprised five lists of five names and five lists of five black-and-white pictures of men's faces. For the task used with the Canadian participants, 25 surnames were selected from the Government's statistical database (Institut de la statistique du Québec, 2002). For the task used with the Belgian participants, names were selected from a Belgian dictionary of surnames (Herbillon & Germain, 1996). Each list comprised three common, one moderately common and one uncommon name (defined according to each database). Names were all six letters long. For the Canadian names, 21 did not begin with the same letters; two pairs had the same first letter in common.

TABLE 2
Neuropsychological profile of the AD group

Neuro psychological testing	*AD participants*															
	1	*2*	*3*	*4*	*5*	*6*	*7*	*8*	*9*	*10*	*11*	*12*	*13*	*14*	*15*	*Mean (SD)*
Age (years of schooling)	84 (18)	68 (9)	85 (8)	72 (12)	76 (12)	76 (8)	58 (11)	72 (7)	79 (7)	73 (11)	76 (8)	76 (7)	74 (19)	73 (9)	56 (11)	73.3 (10.5)
MMSE	22	23	24	24	27	26	19	24	22	25	24	26	29	24	16	23.7 ± 3.2
Working memory																
Digit span forward/ backward	7/5	5/ −	6/3	6/4	5/4	6/4	5/ −	7/4	4/3	6/3	5/5	6/3	8/6	6/2	4/2	10.4 ± 3.1
Block-tapping test (percentile)	30	50	15	12	30	50	10–20	>5	10	10-20	15	10	70	15	10	10-20
Letter-number sequencing	12	–	6	7	9	9	–	3*	2*	10	2*	7	11	7	–	7 ± 3.1
Episodic memory																
Selective cued reminding test:																
Immediate free recall (total/48)	2*	7*	5*	3*	3*	20	6*	0*	3*	14*	2*	4*	12	8*	5*	6.3 ± 5.4*
Immediate cued recall (total/48)	9	26	9	18	6	39	13	5*	18	16	8	14	21	19	11	15.9 ± 9.2
Delayed free recall (total/16)	0*	1*	0*	0*	0*	10	3*	0*	0*	9	0*	1*	5*	3*	2*	2.3 ± 3.3*
Delayed cued recall (total/16)	2	6*	1*	4	0*	14	8*	0*	8	4	3	7	6	5	1*	4.7 ± 3.8
Delayed recognition (total/16)	13*	12*	8*	13	5*	16	13*	13*	16	15	10*	16	14*	15*	12*	12.7 ± 3.1*

Recognition Memory Test (faces) (percentile)	<5	–	<5	<5	10	10	–	<5	25	<5	<5	25	10	5–10	–	<5
Language and semantic memory																
Verbal Fluency (number of words):																
Letter (P)	20	11	10	19	14	10	16	7*	7*	12	20	12	14	4*	2*	11.9 ± 5.5
Semantic	V = 6*	A = 17*	V = 9	V = 12	A = 13*	A = 23	A = 26	T = 7	V = 7	A = 14	V = 12	V = 10	V = 11	V = 7*	A = 5*	–
Association match task (/30)	29	–	28	30	30	–	–	23	27	26	30	27	30	–	28	27.8 ± 2.1
Perception																
Benton's face discrimination	B	–	N	N	N	SI	–	MI	N	N	N	N	N	N	MI	Average
Executive functions																
Stroop:																
Colour naming (errors/ corrected errors)	74″ (0/2)	69″	87′ (2/0)	67″ (0/1)	14″	68″	115″ (1/0)	111″* (1/1)	120″* (0/1)	89″	NT	95″	64″ (0/2)	95″ (1/0)	108″* (9*/0)	84 ± 27.7
Reading (errors/corrected errors)	44″	48″	68″ (0/1)	46″	18″	46″	63″	65″ (2/0)	56″	73″* (2/0)	NT	74″	52″ (0/1)	58″	112″* (6/2)	58.4 ± 21
Interference (errors/ corrected errors)	179″ (1/1)	117″ (2/0)	188″ (61*/0)	156″* (63*/2)	30″	117″	193″ (8*/0)	NT	241″ (9*/7*)	660″* (12/13 *)	NT	231″ (1/0)	119″ (0/4)*	335* (12*/0)	NT	213.8 ± 160.1
Trail Making Test:																
Trail A	55″	57″	159″*	62″	33″	54″	88″	94″*	128″*	79″*	44″	97″*	38″	41″	NT	87.2 ± 65.2*
Test B (errors)	103″	242″*	NT	117″ (3)	140″	113″ (1)	320″* (3)	488″*	765″*	329″*	200″	308″*	91″	238″*	NT	265.7 ± 189.6*

(*Table continued*)

TABLE 2 Continued

Neuro psychological testing	*AD participants*															
	1	*2*	*3*	*4*	*5*	*6*	*7*	*8*	*9*	*10*	*11*	*12*	*13*	*14*	*15*	*Mean (SD)*
Automatic and controlled retrieval in episodic memory, Process Dissociation Procedure:																
Controlled processes (%)	56	25	3*	22	34	-	-	3*	6*	28	0*	16	22	31	0*	25 ± 17.4
Automatic processes (%)	21	33	16	40	38	–	–	10	17	30	22	37	44	50	11	23.3 ± 6.9

SD = standard deviation; p = percentile; * = two standard deviation below the mean control performance, adjusted for age and education; V = vegetables; A = animals; T = tools; NT = could not complete task; B = borderline; N = normal; SI = severe impairment; MI = moderate impairment.

Digit span and letter-number sequencing tests – WAIS III (Wechsler, 1997a); Visuo-spatial block tapping test – WMS-III (Wechsler, 1997b); French version of the Selective and Cued Reminding Test (SCRT) (Calisis, Wyns, Van der Linden, & Coyette, 1994); Recognition Memory Test (Warrington, 1984); Verbal fluency – French version (Joanette et al., 1995); Association match task from the Birmingham Object Recognition Battery (BORB) (Riddoch & Humphreys, 1993); Benton's Face Discrimination Test (Benton, Hamsher, Varney, & Spreen, 1983); The Stroop Test (Golden, 1978); The Trail Making Test A and B (Partington & Leiter, 1949; as cited in Spreen & Strauss, 1998); The Process Dissociation Procedure (PDP) (Adam et al., 2005). For the PDP, the results of the AD participants were compared to those of the 15 matched control participants: controlled processes = 42.5 ± 18.2, automatic processes = 17.5 ± 13.4.

For the Belgian names, the 25 names all had a different first letter. None of the names, Canadian and Belgian, had a semantic referent; for example, last names having a semantic meaning (e.g., "Lahaie" – the hedge) or associated with a known personality (e.g., politics, sports). The 25 photographs were selected from a sample of 126 pictures. They were the same in the Canadian and Belgian tasks. The photographed people were all men between 20 and 50 years of age and were without distinctive facial characteristics, e.g., moustache, glasses, orientation, or position. They were selected from the material used by Willems, Adam, and Van der Linden (2002).

Design

This study consisted of a within-subject design in which all participants were exposed to all learning methods. AD patients were seen twice a week, at a two-day interval, over five consecutive weeks for a total of 10 45-minute sessions. The face–name association task was presented to participants under five different learning conditions: three target methods (spaced retrieval, errorless and vanishing cues) and two control methods (trial-and-error). For each participant, each list of face–name associations was randomly assigned to be used with one of the five learning methods. Each method was assessed in one session divided into three parts: (1) a learning phase of 30 minutes; (2) a two-minute break; (3) a free recall, cued recall and recognition phase. Thus, for each method, there was the same number of sessions and the same amount of time allotted to practice and breaks. Two weeks after each learning session, a delayed free recall, cued recall and recognition phase was administered. The two-week period was based on the fact that it took two weeks for all participants to be exposed to the five methods. Each method was assessed at a two-day interval for a total of five delayed recall sessions.

Control participants were seen once or twice a week at their convenience. Delayed testing was administered in one session, instead of five, one week after application of the last learning method. This was done in order to decrease the time requirements for these participants and thus the attrition rate.

Procedure

Participants were told that they were going to see pictures of five people with their associated names. They were asked to learn the associations between faces and names because they would have to retrieve them at the end of the session. During the learning phase, different instructions were given to the subjects when the correct answer was not retrieved, depending on which method they were exposed to. In all target methods and one control method (implicit), implicit memory test instructions were used

(Graf & Schacter, 1985). The participants were told to give the first answer that came to mind, even if it seemed like a guess, and no reference was made to the preceding learning trials. For the other control method (explicit), the participants were explicitly asked to try to remember the names presented earlier. The explicit control method was always presented first to all participants. This procedure was followed in order to avoid having the participants always give the first answer that came to mind (perseveration of the instruction), even during the explicit control method. The other four methods were presented in a counterbalanced order. During the learning and test phases, the experimenter recorded the answers given by the subject and the response time.

Specific procedure for each learning method

Spaced retrieval. One face–name association was first presented for 5 seconds. The experimenter gave the following instructions: "Here is a picture of a man, whose name is Mr X. Can you repeat this name and try to remember it?" The name was then hidden, but not the picture, and the participant was asked to give its associated name (0 seconds recall interval). The picture was presented again 10 seconds later to the participant, who once more had to recall its associated name upon being asked: "What is the name of this man?" No reference was made to the preceding trial and, if the participants could not remember the name, they were asked to give the first name that came into their mind. This procedure was repeated at increasing time recall intervals of 0 seconds, 10 seconds, 20 seconds, 30 seconds, 1 minute, 1(1/2), 2, 2(1/2), 3, 3(1/2), 4, 4(1/2) and 5 minutes. If the participant made a mistake, the experimenter gave the right answer and returned to the last successful recall interval, after which the series of time intervals was continued. If the participant did not know the answer, they were instructed to give the first answer that comes to mind. The five associations were learned in the same session. A new association was introduced every minute, the first association beginning at time 0, the second association at 1 minute, the third at 2 minutes, and so on (see Figure 1), each one with its own recall progression. The session ended after 30 minutes or when the success criterion was reached for each association (5-minute recall interval).

Errorless learning. A first face–name association was presented for five seconds with the following instructions: "Here is a picture of a man, whose name is Mr X. Can you repeat this name and try to remember it?" The picture and name were both hidden, then the picture was presented again, immediately. The participant was asked to give its associated name with the same instructions as the spaced retrieval method. This procedure was chosen in order to apply an errorless method but with an effortful component. The classic errorless method, in which patients are presented with the

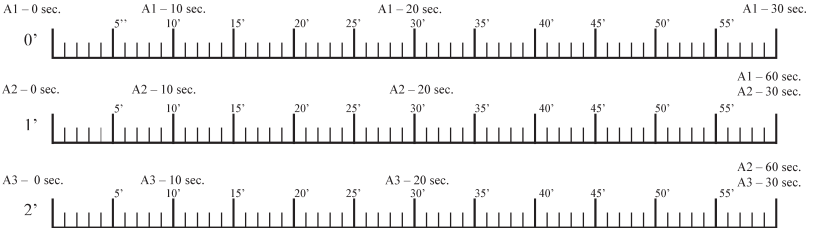


Figure 1. Example of recall intervals with the spaced retrieval method. Each line represents 1 minute broken down into 60 seconds. This figure represents the first 3 minutes of the intervention with example of recall intervals reached by association 1, 2 and 3 (without error). A1 = association 1; A2 = association 2; A3 = association 3; sec. = seconds.

answer and must repeat this answer immediately, was criticised by some authors since it is a very passive way to learn (Jones & Eayrs, 1992; Wilson & Evans, 1996). We expected very few mistakes with this procedure since the information is presented and retrieved immediately by the participants. The other associations were presented in the same way, in three cycles of three presentations and in random order. The participants had 20 seconds to give an answer. If the participant made a mistake or could not give the name, the experimenter gave the right answer. If the participant did not know the answer, they were instructed to give the first answer that comes to mind. Each name was repeated nine times in each session. Each cycle was separated by a 30-second break.

Vanishing cues. The five face–name associations were each presented for 5 seconds. Each time, the experimenter gave the following instructions: "Here is a picture of a man, whose name is Mr X. Can you repeat this name and try to remember it?" The picture was represented with the first five letters of the name. Participants were asked to recall the complete name with the implicit memory instructions. The other associations were presented in random order, following the same procedure. In the following presentations, the number of letters was progressively decreased (e.g., FORTIN, FORTI, FORT, etc.). When the participants made a mistake, a letter was added, but not read by the experimenter, until they were able to recall the target name. If the participant did not know the answer, they were instructed to give the first answer that comes to mind. In the next trial, the number of letter cues given was one less than the number required in the preceding trial, and so on until the word was recalled without any cues. The participants had 20 seconds to give an answer. The procedure ended after three consecutive successful trials with no letter cues or after 30 minutes. This procedure was based on a backward chaining approach of vanishing cues, as proposed by Fontaine (1995), to avoid error production as much as possible.

Trial-and-error method with explicit memory task instructions. The five face–name associations were each presented for 5 seconds. For each picture, the experimenter gave the following instructions: "Here is a picture of a man, whose name is Mr X. Can you repeat this name and try to remember it?" One picture was then presented again and the participant was asked to give its associated name. The experimenter emphasised that this picture and its associated name had been presented before and explicitly asked the participant to remember the correct answer. The participants had 20 seconds to give an answer. If they made a mistake, the experimenter gave the right answer. The same procedure was repeated with the other associations (in random order) in three cycles of three presentations, separated by a 30-second break, for a total of nine presentations.

Trial-and-error method with implicit memory task instructions. This method was almost identical to the previous control method and comprised the same trial-and-error component. The difference between the two was in the instructions given to the participants. If they did not know the answer, they were administered the same implicit memory instructions as those given in the three target methods.

Test phase

The test phase was the same for each method and consisted of free recall, cued recall and recognition of the face–name associations. Under *free recall*, each picture was presented and the participants were asked to recall its associated name in response to explicit instructions. If they could not give the right answer after 20 seconds, they were cued with the first letter of the name. If they still could not give the right answer after another 20 seconds, a second letter was added, and so on, until they could give the right answer (*cued recall*). After a 2-minute break, the five names were presented with 10 pictures comprising the 5 learning faces and 5 distractors (*recognition*). The participant had to match the names with the right faces. The 2-minute break was chosen to ensure that the information was not maintained in working memory (Grober & Buschke, 1987). The same free recall, cued recall and recognition tests were administered two weeks later.

Statistical analyses

Since the variables did not follow a normal distribution (Shapiro-Wilk, ps $< .05$), non-parametric tests were used. Objective 1 (specific efficacy of each method) was explored by means of Wilcoxon signed rank tests (Z) to compare the group performance on each method individually with a performance criterion. The AD group performances were also compared to the control group performances using Mann-Whitney U tests. Objective 2, comparison of

the five methods, was explored with a Friedman test (χ^2) which compares the performances of the participants on each method (repeated measures). Two-by-two analyses were also performed with Wilcoxon signed rank tests. Objective 3 (error production) was examined with Mann-Whitney *U* tests to compare the performances of the participants on free recall between face–name associations which led to a greater number of mistakes and associations which led to fewer mistakes. For objective 4, the participants' performances on the methods using implicit instructions were combined and compared to their performances on explicit trial-and-error by means of Wilcoxon signed rank tests. For objective 5, Spearman correlations were performed to examine the association between the group's cognitive profile and its performance on the five methods. We also compared, for each method, the cognitive profile of the five best and five worst performers with Mann-Whitney *U* tests. Finally, modified *t* tests for single case studies (Crawford & Howell, 1998) were also carried out to compare each AD participant's performances individually with the control group and identify the method for which the greatest number of AD subjects obtained a performance as good as the controls. Modified *t* tests measured whether a single observation was significantly different from the mean of a small control sample.

RESULTS

Descriptive analyses of the learning sessions

The mean number of item presentations and the proportion of errors produced with each method during the learning sessions are presented in Table 3. In the AD group, spaced retrieval led to the greatest number of item presentations

TABLE 3
Learning session analysis of the five methods (AD and Control groups)

	Mean number of presentations for each association		*Proportion of errors produced by trial*	
Methods	*AD*	*Controls*	*AD*	*Controls*
Spaced retrieval	22 ± 2	15.3 ± 2.6	2.1 ± .56	.75 ± .78
Vanishing cues	10.7 ± 2.5	8.9 ± .81	2.4 ± .72	1.4 ± .87
Errorless	9	9	.40 ± .32	.10 ± .16
Trial-and-error explicit	9	9	4.2 ± .59	1.1 ± 1.0
Trial-and-error implicit	9	9	4.3 ± .59	1.4 ± .88
Target methods			*1.4*	*.80*
Control methods			*4.2*	*1.3*

during the learning trial. The other four methods led to a comparable number of presentations (9 or 10 per session). As for error production, the AD group produced significantly more mistakes than the control group (Mann-Whitney U test $ps < .01$). The proportion of errors was computed by dividing the total number of errors produced during the session by the number of item presentations. The proportion of errors was computed since the number of item presentations was different between the five methods. The proportion of errors thus gave us a comparative measure. The errorless method produced the fewest proportion of mistakes followed by spaced retrieval and vanishing cues. With the two control methods, the AD participants made mistakes on almost every association in each learning trial. In fact, the AD group produced a significantly higher proportion of errors with the control methods than with the target methods (Wilcoxon's $Z = -3.41$; $p < .001$). Finally, the main type of errors produced, within-list mistakes, was similar for all five learning methods (see Figure 2).

Specific efficacy of the learning methods on immediate recall

The AD participants' performances on immediate free recall and recognition are presented in Table 4. For cued recall, the performances were computed following Buschke et al.'s (1999) procedure which consists of calculating a total score combining free recall and cued recall. Since letter cues (maximum of 6 letters) were added only if free recall failed, 6 points were

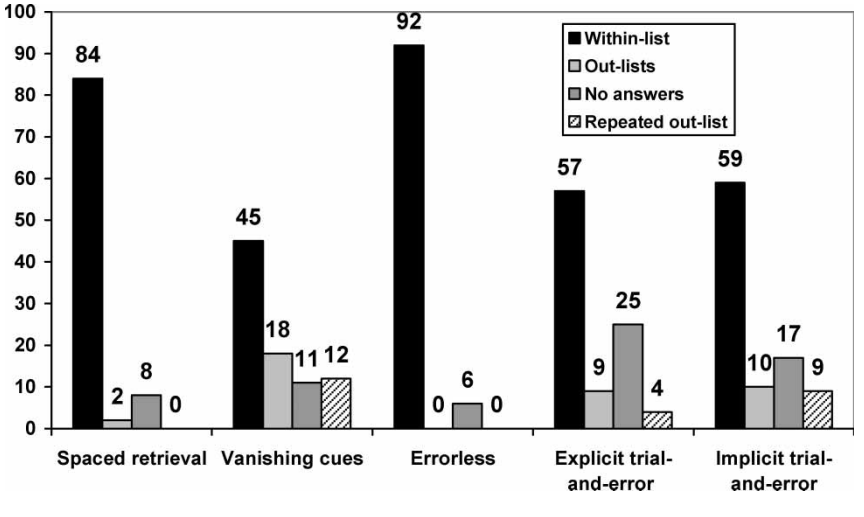


Figure 2. Main types of errors produced by AD participants for each method during the learning phases (results expressed in percentages).

TABLE 4
Test performances of the AD and Control groups on immediate and delayed recall

	Learning methods				
Performance measures	*Spaced retrieval*	*Vanishing cues*	*Errorless*	*Explicit trial-and-error*	*Implicit trial-and-error*
Immediate					
Free recall (/5)					
AD: Mean ± *SD*	1.5 ± 1.5	1.1 ± 1.4	.80 ± .86	1.3 ± 1.2	.60 ± .74
Max	4	5	3	3	2
Controls: Mean ± *SD*	4.5 ± 1.1	4.5 ± 1.1	3.9 ± 1.1	4.5 ± .83	4.3 ± 1.0
Total score combining free and cued recall (/30):					
AD: Mean ± *SD*	22 ± 6.5	22.7 ± 4.9	18.9 ± 5.9	21.9 ± 5.4	21.5 ± 3.2
Controls: Mean ± *SD*	29.4 ± 1.5	29.5 ± 1.1	28.7 ± 1.3	29.5 ± .83	29.3 ± 1.0
Recognition (/5)					
AD: Mean ± *SD*	1.8 ± 1.7	1.3 ± 1.5	1.6 ± 1.0	1.6 ± 1.6	1.0 ± 1.7
Max	5	4	3	5	5
Controls: Mean ± *SD*	4.7 ± 1.0	4.3 ± 1.4	4.9 ± .51	4.6 ± 1.1	4.6 ± .91
Delayed					
Free recall (/5)					
AD: Mean ± *SD*	.10 ± .29	.30 ± .45	0 ± 0	0 ± 0	0 ± 0
Proportion of associations retained	7%	27%	0%	0%	0%
Controls: Mean ± *SD*	.40 ± .92	.50 ± .69	.40 ± 1.2	.50 ± .67	.40 ± .82
Total score combining free and cued recall (/30):					
AD: Mean ± *SD*	22 ± 6.5	22.7 ± 4.9	18.9 ± 5.9	21.9 ± 5.4	21.5 ± 3.2
Controls: Mean ± *SD*	29.4 ±1.54	29.5 ±1.12	28.7 ±1.34	29.5 ± .83	29.3 ± 1.05
Recognition (/5)					
AD: Mean ± *SD*	.80 ± 1.1	1.4 ± 1.3	.70 ± .65	1.7 ± 1.6	.80 ± 1.1
Controls: Mean ± *SD*	2.7 ± 2.0	3.5 ± 1.8	2.1 ± 1.8	2.7 ± 2.0	2.7 ± 1.8

SD = standard deviation; Max = Maximum score obtained by the best participant.

given when a name was correctly free-recalled, 5 points were given when the name was not free-recalled but correctly recalled with the help of 1 letter, 4 points with the help of 2 letters, and so on for a possible total of 30 points (5 names – maximum of 6 points for each). This total score is presented in Table 4.

As a whole, controls had better performances than AD patients on each immediate test (Mann-Whitney *U* tests: free recall $ps < .001$; total score combining free and cued recall $ps < .001$; recognition $ps < .001$). In order to assess the efficacy of each method for the AD group, the performances obtained by AD participants in the test phases were compared with a performance criterion

of 0 considered a failure. Results showed that the five learning methods were efficient (free recall: Wilcoxon's Zs between -2.4 and -2.8, ps $< .005$; total score combining free and cued recall: Wilcoxon's Zs of -3.41, ps $< .001$; recognition: Wilcoxon's Zs between -2 and -3.1; ps $< .01$).

Comparisons of the learning methods on immediate recall

The comparison of the five learning methods for the AD group showed that there were no significant differences between them, neither on free recall (Friedman's $\chi^2 = 6.8$; $p = .15$), on total score combining free and cued recall (Friedman's $\chi^2 = 5.3$; $p = .25$), on recognition (Friedman's $\chi^2 = 2.3$; $p = .68$), nor on mean response time computed for correct answers only (Wilcoxon's Zs between -0.10 and -2.2; ps $> .005$; Bonferroni correction $0.05/10$). For control participants, there were no significant differences between their performances on the five methods on free recall (Friedman's $\chi^2 = 7.7$; $p = .10$), on the total score combining free and cued recall (Friedman's $\chi^2 = 7.7$; $p = .10$) or on recognition (Friedman's $\chi^2 = .20$; $p = .70$).

Comparison of the learning methods on delayed recall

To see if some methods led to better maintenance of learning over time, the differences between immediate and delayed performances were compared. Results for the AD group are presented in Table 4. For this group, performances were poorer on delayed free recall than on immediate free recall (Wilcoxon's Z, ps $< .01$). The participants retained between 0% and 27% of the associations. On the total score combining free and cued recall, AD participants recalled more associations during immediate than during delayed recall (Wilcoxon's Z, ps $< .005$). Finally, on recognition, AD performances were not different on immediate and delayed recall (Wilcoxon's Zs between 0 and -1.4; ps $> .20$), except for the errorless method, which was better during immediate testing (Wilcoxon's $Z = -2.4$; $p = .02$).

Comparison between the five methods showed a significant difference between AD performances when tested on free recall (Friedman's $\chi^2 = 9.7$, $p = .046$), with a trend towards a significant advantage for vanishing cues over the other methods (Wilcoxon's Z, ps $= .083$). There were no differences between the five methods on total score combining free and cued recall and on recognition (Wilcoxon's Z, ps $> .10$).

For the control group, performances were also lower two weeks later when measured with free recall and with the total score combining free and cued recall (Friedman's χ^2, ps $< .005$). They retained between 1% and 10% of the associations when tested on free recall. For delayed recognition, performances of the control group were also poorest during delayed recall (Friedman's χ^2, ps $< .027$), except for vanishing cues and explicit

trial-and-error, where they obtained similar results between delayed and immediate recognition. No significant differences were observed between the five methods (Friedman's χ^2, $ps > .50$).

Influence of error production and type of instructions

To assess if errors produced might have influenced the AD participants' results on the tests, we first computed the number of errors produced during the learning phase by each participant individually, for each face–name association. Then, for each method, free recall performances obtained with the face–name associations leading to the greatest number of mistakes was compared with the face–name associations that led to the fewest mistakes. For the two trial-and-error methods, the associations that led to fewer mistakes during the learning phase were better retrieved during free recall (explicit trial and error: Mann-Whitney $U = 233.0$, $p = .002$; implicit trial and error: Mann-Whitney $U = 274.0$; $p = .001$). There were no differences for the target methods (Mann-Whitney U, $ps > .30$).

A final analysis was carried out to assess if the four methods using implicit memory test instructions optimised learning better than the explicit trial-and-error method. The participants' performances on the four implicit methods were combined and compared with their performances on the explicit trial-and-error method. In the AD group, the type of instructions did not have any impact on performances during the test phase (Wilcoxon's, Z, $ps > .30$). The same results were obtained for control participants (Wilcoxon's, Z, $ps > .10$).

Correlation between AD participants' performances and their cognitive profile

The purpose of this analysis was to explore the relationship between the AD group's performances on free recalls and the neuropsychological battery (the battery is presented in Table 2) in order to characterise the cognitive functions associated with good performances on the five methods. Bivariate Spearman correlations were carried out between the following cognitive domains and the AD group performances on the total score of the five methods – Bonferroni correction $0.05/11$; $p \leq .005$: (1) Global functioning, measured by the Dementia Rating Scale; (2) Episodic memory functioning, measured by the Selective and Cued Reminding Test (SCRT) total of immediate recalls (free + cued recalls), the SCRT total of delayed recalls (free + cued recalls), and the controlled processes score on the Process Dissociation Procedure (PDP); (3) Inhibition, measured by the Stroop interference score; and (4) Automatic retrieval, measured by the PDP's automatic processes score. Those variables were chosen based on the hypothesis that the learning methods would be most beneficial for patients with more severe memory

deficits, who should use their automatic retrieval more. We also hypothesised that the methods should potentially be more beneficial for patients with inhibition deficits, who tended to produce more errors (Amieva et al., 1998).

The learning methods were not correlated with global cognitive functioning (*r*s between .03 and .30; *p*s $>$ .05). AD participants with better episodic memory obtained better results on explicit trial-and-error and vanishing cues methods (*r*s between .69 and .80; *p*s $<$ .004). Also, participants with the greatest inhibition deficits obtained better results on the vanishing cues method ($r = .60$; $p < .005$). The automatic retrieval measure was not correlated with the participants' performances.

Case study analyses

In order to better identify the characteristics of the AD participants who could benefit most from the methods and those who may not, case study analyses were done on the basis of the total score combining free and cued recall. The total score was chosen instead of free recall since it allowed us to identify more subtle differences between the participants. The cognitive profile of the five best performers was compared with the cognitive profile of the five worst performers on each learning method and for the following neuropsychological tests (Mann-Whitney *U* tests): (1) General cognitive function (Mattis Dementia Rating Scale), (2) Episodic memory: SCRT total immediate recall and total delayed recall and the PDP's controlled processes scores; (3) Inhibition, measured by the Stroop interference score; and (4) Automatic retrieval, measured by the PDP's automatic process score.

For the five methods, the best performers had better remaining episodic memory as measured by the SCRT total immediate recall than the worst performers (Mann-Whitney *U*, *p*s $<$ .059 on SCRT immediate recall and *p*s $<$ .028 for SCRT delayed recall). For the explicit trial-and-error method, the five best participants had better general cognitive function as measured by the Mattis dementia rating scale (Mann-Whitney $U = 2.0$; $p = .028$) and relied more on their automatic processes (Mann-Whitney $U = .50$; $p = .019$). For the other learning methods, there were no other significant differences between the best and worst performers. Finally, the five best and five worst performers were compared on the number of mistakes they produced during learning, for each method. For the trial-and-error methods, the five best performers tended to produce fewer mistakes than the five worst performers (explicit trial-and-error: Mann-Whitney $U = 3.5$; $p = .058$; implicit trial-and-error: Mann-Whitney $U = 4.0$; $p = .069$). For the other methods, there were no differences on the number of errors produced between the two subgroups of participants (Mann-Whitney *U p*s $>$ .25).

A second case study analysis aimed at identifying the method for which the greatest number of subjects obtained a performance as good as the controls. For

each method, the performance on immediate free recall of each AD subject was compared with control participants by means of modified *t* tests for single case studies (Crawford & Howell, 1998). AD subjects who did not obtain a performance different from the controls were considered the best participants. For the errorless condition, subject 10 obtained a performance at free recall not significantly different from the control group ($t = -0.79$, $p = .22$). He obtained a mean of three correct associations out of five. For vanishing cues, subject 10 again performed as well as the controls ($t = 0.39$; $p = .35$) and obtained a performance of five correct associations. Spaced retrieval was the method for which the greatest number of AD patients obtained a comparable performance to the controls: subjects 1 and 10 had three correct answers and subjects 5 and 13 had four correct answers. These performances were not significantly different from the control group ($ps > .10$). Finally, no subject obtained a performance as good as the control group for the two control methods.

To assess if subjects 1, 5, 10 and 13 were significantly different from the rest of the AD group, *Z* scores were computed between each score of the four subjects individually on neuropsychological tests and the mean score of the AD group. *Z* scores of more than 2 and less than −2 (inclusive) were considered significantly different from the group. Subjects 1 and 5 had worse episodic memory deficits than the group (subject 1: immediate free recall of the SCRT, $Z = -2.15$; subject 5 delayed recognition of the SCRT, $Z = -2.5$). Subject 10 had better episodic memory (delayed free recall of the SCRT; $Z = 2.03$) but significantly worse inhibition deficit (interference condition of the Stroop test; $Z = -2.8$). Finally, subject 13 did not stand apart from the group except that he obtained a high score on the automatic processes estimation of the PDP ($Z = 3$).

A final analysis was carried out to assess if those subjects had specific patterns of learning in terms of the number of presentations and proportion of errors produced. Again, *Z* scores were computed between each subject's performances and the AD group. *Z* scores of more than 2 and less than −2 (inclusive) were considered significantly different. Only one small difference was observed: subject 1 had more item presentations with the spaced retrieval method than the group (37 presentations; $Z = 2.3$).

GENERAL DISCUSSION

This study investigated the performance of early AD participants and controls on five different learning methods: spaced retrieval, errorless learning, vanishing cues, and two trial-and-error methods (with explicit or implicit memory test instructions).

The first and second objectives of this study were to explore the efficacy of each method and to compare them with each other. In the AD group, the

results showed that the five methods led to significant and similar learning of face–name associations as measured by free recall, total score combining free and cued recall, and recognition. The AD group's performances were, however, inferior to the controls and not maintained in delayed recalls. The specific efficacy of the three target methods observed during immediate recall is in agreement with previous findings showing the potential of each method in early AD (Camp et al., 1996b; Clare et al., 2000; Fontaine, 1995; Winter & Hunkin, 1999). The observation that the trial-and-error methods were also effective was unexpected but is in line with recent findings (Haslam et al., 2006; Metzler-Baddeley & Snowden, 2005; Ruis & Kessels, 2005), which showed that AD participants could learn with both errorless and trial-and-error learning conditions. Our AD group's results are also in line with Hochhalter et al.'s (2005) study. These authors compared spaced retrieval with different types of practice schedules and found no advantage of one method over the other.

As for the results obtained in the control participants, we observed that, in general, they obtained similar results with the five learning methods. These results are not in accordance with previous studies (Baddeley & Wilson, 1994; Kessels & de Haan, 2003). In the Kessels and de Haan (2003) and Baddeley and Wilson (1994) studies, errorless learning was superior to trial-and-error. As for Fontaine's (1995) study, results showed that vanishing cues and trial-and-error were equally effective. Nevertheless, it is possible that we would have obtained different results with more stimuli and this hypothesis could be part of a more extensive study focusing on the role of errors for new learning in unimpaired aged people.

The third objective of this study was to explore if error production might be an important factor to promote new learning in early AD. We observed that all methods led to error production, although the trial-and-error methods led to twice as many mistakes produced during the learning phase as the target methods. The target methods were, consequently, not totally error-free. In general, error production was not strongly correlated with performances on free recall and on total score combining free and cued recalls. Case study analyses were in line with the group results and showed that the best AD participants did not produce fewer mistakes than the rest of the group. These results raise questions regarding the role of errors in promoting learning in this population. It may be that the limitation of error production is not the main factor explaining why AD subjects can learn with these methods. The same conclusion was suggested by Hochhalter and collaborators (2005). Their study, carried out with four patients, showed that spaced retrieval did not reduce error production during training and was in fact one of the conditions leading to the greatest number of mistakes. Furthermore, all learning conditions, like spaced retrieval, massed repetition and expanding rehearsal, led to equivalent results in terms of new learning of a verbal task. As for

Haslam and colleagues (2006), they showed that errorless learning tended to have an advantage over trial-and-error learning when their participants tried to recall specific details; for example, the name of the person was better recalled with errorless learning than its familiarity. It may be that methods that limit error production are effective for certain types of material only and this hypothesis should be explored in future studies carried out with more participants.

In addition to the role of errors in learning, the role of implicit memory instructions in the methods' efficacy was also explored. We observed that, regardless of the type of instructions used, implicit or explicit, the AD participants obtained equivalent results. Moreover, as a whole, participants with better episodic memory, i.e., those who are prone to use their residual explicit memory capacities, tended to benefit more from the different learning methods. The case study analyses however showed less clear results regarding remaining episodic memory capacities: two out of the four best participants had worse episodic memory deficits than the rest of the group. Our results thus raise questions about the importance of optimising implicit (automatic) retrieval to enhance learning in AD. To clarify our results, it would be interesting to measure the contribution of implicit memory by using a sensitive measure of this kind of retrieval, such as the Process Dissociation Procedure (PDP) (Adam, Van der Linden, Collette, Lemauvais, & Salmon, 2005; Jacoby, 1991). This procedure is based on the hypothesis that retrieval in episodic memory is influenced by explicit (controlled) processes and by implicit (automatic) processes (Jacoby, 1991). From this perspective, Jacoby (1991) developed a measure to separate the contributions of controlled and automatic processes to the performance of a subject on a single memory task, the Process Dissociation Procedure (PDP). This procedure comprises two conditions, one in which controlled and automatic processes act in concert to influence performance in the same way and another condition in which controlled and automatic processes have opposing effects. The contribution of each process can then be determined with simple equations. The PDP, which is more suited to distinguish between implicit (automatic) and explicit (controlled) processes than classic memory tests, could help to identify if the stimuli learned with the different methods are primarily accessed by controlled or by automatic processes and if there are differences between the learning methods in this regard (see Anderson & Craik, 2006, and Kessels, Boekhorst, & Postma, 2005, for a similar application of the PDP on errorless learning in healthy younger and older adults).

Overall, therefore, our results suggest that AD patients can learn new information regardless of the number of errors produced and the type of retrieval optimised during learning. This means that other optimising factors may be important to promote new learning in this population and could be shared by all methods. One of the common aspects of the procedures used for the five

methods examined in this study is that the participants had to produce an answer in each trial. Learning was never passive, like reading or repeating an answer provided by the experimenter. This active role of the participants during the learning phases shares similar characteristics with the self-generated response effect. This phenomenon refers to the demonstration that responses generated by the subjects themselves are remembered better than when they are externally provided by the experimenter (Slamecka & Graf, 1978). It refers not just to "learning by doing" but also to the participant's active involvement in the learning process (Slamecka & Graf, 1978). In AD, this phenomenon has been explored in learning lists of words and results showed that the participants tended to perform better when they generated answers by themselves (Barrett, Crucian, Schwartz, & Heilman, 2000; Dick, Kean, & Sands, 1989). The generation effect may be explained by the fact that the act of generating an answer is a recall of the stimulus and thus reinforces the "access road" to this stimulus (Slamecka & Graf, 1978). This phenomenon shares common aspects with the retrieval effect, which refers to the long-term superiority of training done by successive recovery of information compared to encoding carried out on several occasions (Bjork, 1988). The retrieval of an item, like its generation, could reinforce the "access road" to its mnesic trace. Since this "road" is used on several occasions, the item becomes increasingly accessible and provides another encoding each time it is retrieved (Baddeley, 1994; Bird, 2001). It might be that what is important in new learning in AD is this act of generating or retrieving an answer. In fact, our case studies indicate that spaced retrieval was the method for which the greatest number of participants obtained results as good as the controls. In the normal subjects, the efficacy of spaced retrieval was attributed in part to the retrieval effect (Landauer & Bjork, 1978). The role of this effect in optimising new learning in AD, which has not been studied to our knowledge, could be explored more specifically in the future in addition to the role of the generation effect.

This being said, given the relatively small number of subjects involved in our study, interpretations concerning these results must be viewed with caution. Different results on recalls may be observed with more patients or with more learning sessions. More learning sessions might also have enhanced long-term retention. Regarding our procedures, learning one association at a time, instead of the entire list of stimuli in the same session, may decrease the amount of intra-list interference and enhance learning. Finally, controlling for the amount of item presentations (ensuring that each method has the same amount of item presentations), may produce different results.

In conclusion, the five methods were efficient for face–name learning in the present study. Error production and implicit memory did not clearly enhance learning and this suggests that other factors, such as self-generated responses or the retrieval effect, could optimise learning in patients with AD. From a clinical point of view, we cannot recommend one method over

the others, but our results confirm Camp's (1989) suggestion to select a limited amount of information since our AD patients learned a mean of one or two associations per session. Furthermore, one learning session is not enough to achieve long-term retention and many trials should be done to ensure the clinical relevance of these methods. It also seems that, generally, AD patients with better episodic memory or with greater inhibition deficits may benefit more from the different methods but the cognitive profile of the best performers should be clarified in future studies. Finally, little attention has been paid to the real potential of these methods in improving autonomy in daily living, although they have been applied to potentially relevant tasks. This is an important aspect for future clinical research and should be explored in further studies.

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